

Mod5 counter

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1 Synchronous Mod-5 Counter

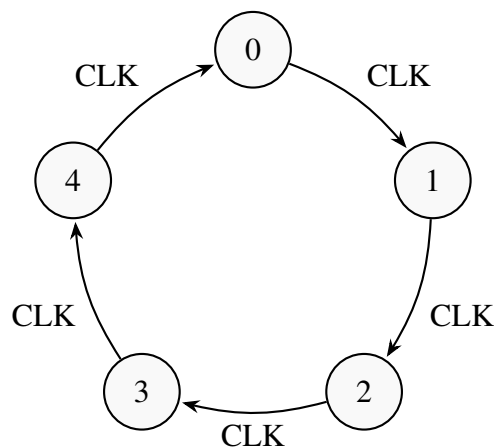
1.1 Aim

To prepare a mod-5 Synchronous Counter using JK Flip-Flops. We should be able to see 0 to 4 and after it resets to 0. To avoid any random issues like locking of states set state 5, 6, 7 to 0.

1.2 Apparatus

Bread Board, 2 7476 IC's (2 JK flip-flops), 1 7408 IC (and gate),
1 7402 IC (Nor gate), 3 LED's, 3 resistors

1.3 Theory



State transition diagram of mod-5 counter

The counter uses three JK flip-flops (Q_2, Q_1, Q_0). The normal counting sequence is 0 to 4. To ensure stability, the unused states (5, 6, and 7) are forced to transition to 0 (000) on the next clock pulse.

Current State			Flip-Flop Inputs						Next State		
Q_2	Q_1	Q_0	J_2	K_2	J_1	K_1	J_0	K_0	Q_2	Q_1	Q_0
0	0	0	0	X	0	X	1	X	0	0	1
0	0	1	0	X	1	X	X	1	0	1	0
0	1	0	0	X	X	0	1	X	0	1	1
0	1	1	1	X	X	1	X	1	1	0	0
1	0	0	X	1	0	X	0	X	0	0	0
1	0	1	X	1	0	X	X	1	0	0	0
1	1	0	X	1	X	1	0	X	0	0	0
1	1	1	X	1	X	1	X	1	0	0	0

Karnaugh Maps

Below are the K-maps for the J and K inputs of each flip-flop. We use Q_2 for the rows and $Q_1 Q_0$ for the columns.

Flip-Flop 2 (MSB)

K-map for J_2 :

$Q_2 \setminus Q_1 Q_0$	00	01	11	10
0	0	0	1	0
1	X	X	X	X

Equation: $J_2 = Q_1 Q_0$

K-map for K_2 :

$Q_2 \setminus Q_1 Q_0$	00	01	11	10
0	X	X	X	X
1	1	1	1	1

Equation: $K_2 = 1$

Flip-Flop 1

K-map for J_1 :

$Q_2 \setminus Q_1 Q_0$	00	01	11	10
0	0	1	X	X
1	0	0	X	X

Equation: $J_1 = \overline{Q_2} Q_0$

K-map for K_1 :

$Q_2 \setminus Q_1 Q_0$	00	01	11	10
0	X	X	1	0
1	X	X	1	1

Equation: $K_1 = Q_2 + Q_0$

Flip-Flop 0 (LSB)

K-map for J_0 :

$Q_2 \setminus Q_1 Q_0$	00	01	11	10
0	1	X	X	1
1	0	X	X	0

Equation: $J_0 = \overline{Q_2}$

Final Logic:

K-map for K_0 :

$Q_2 \setminus Q_1 Q_0$	00	01	11	10
0	X	1	1	X
1	X	1	1	X

Equation: $K_0 = 1$

$$J_2 = Q_1 \cdot Q_0 \quad (1)$$

$$K_2 = 1 \quad (2)$$

$$J_1 = \overline{Q_2} \cdot Q_0 \quad (3)$$

$$K_1 = Q_2 + Q_0 \quad (4)$$

$$J_0 = \overline{Q_2} \quad (5)$$

$$K_0 = 1 \quad (6)$$

1.4 Connections

- This is the most important part in the experiment as even a single faulty connection will lead to wrong output.
- The circuit requires three flip-flops. We will use two 74HC76 ICs (Dual JK Flip-Flops).
- Let IC1 contain Q_0 and Q_1 . Let IC2 contain Q_2 .
- We also use a 74HC08 (Quad AND) as IC3 and a 74HC02 (Quad NOR) as IC4.

Step 1: Power, Ground

- **VCC (+5V):** Connect to IC1(Pin 5), IC2(Pin 5), IC3(Pin 14), and IC4(Pin 14).
- **Ground (0V):** Connect to IC1(Pin 13), IC2(Pin 13), IC3(Pin 7), and IC4(Pin 7).
- **PRESET and SET:** The 74HC76 has active-low Preset and Set pins. Tie them all to HIGH (+5V) : IC1(Pins 2, 3, 7, 8) and IC2(Pins 2, 3, 7, 8).

Step 2: Synchronous Clock

- Connect the output of the 1.00 Hz Function Generator simultaneously to the clock inputs of all three flip-flops: IC1(Pin 1), IC1(Pin 6), and IC2(Pin 1).

Step 3: LSB Flip-Flop (Q_0) Wiring

- $J_0 = \overline{Q_2}$: Connect IC2(Pin 14) ($\overline{Q_2}$) directly to IC1(Pin 4) (J_0).
- $K_0 = 1$: Connect IC1(Pin 16) (K_0) directly to +5V.

Step 4: Middle Flip-Flop (Q_1) Wiring

- $J_1 = \overline{Q_2} \cdot Q_0$:
 - Connect IC2 Pin 14 ($\overline{Q_2}$) to AND gate IC3 Pin 10.
 - Connect IC1 Pin 15 (Q_0) to AND gate IC3 Pin 9.
 - Connect the AND output IC3 Pin 8 to IC1 Pin 9 (J_1).
- $K_1 = Q_2 + Q_0$: (Using NOR gates to synthesize OR logic)
Use the fact that $Q_2 + Q_0 = \overline{\overline{Q_2} \cdot \overline{Q_0}}$
 - Connect IC2 Pin 14 ($\overline{Q_2}$) to AND gate IC3 Pin 13.
 - Connect IC1 Pin 14 ($\overline{Q_0}$) to AND gate IC3 Pin 12.
 - Connect the AND output IC3 Pin 11 to both the inputs of IC4 i.e Pin 12 and Pin 11.
 - Connect the final output IC4 Pin 13 to IC1 Pin 12 (which is K_1).

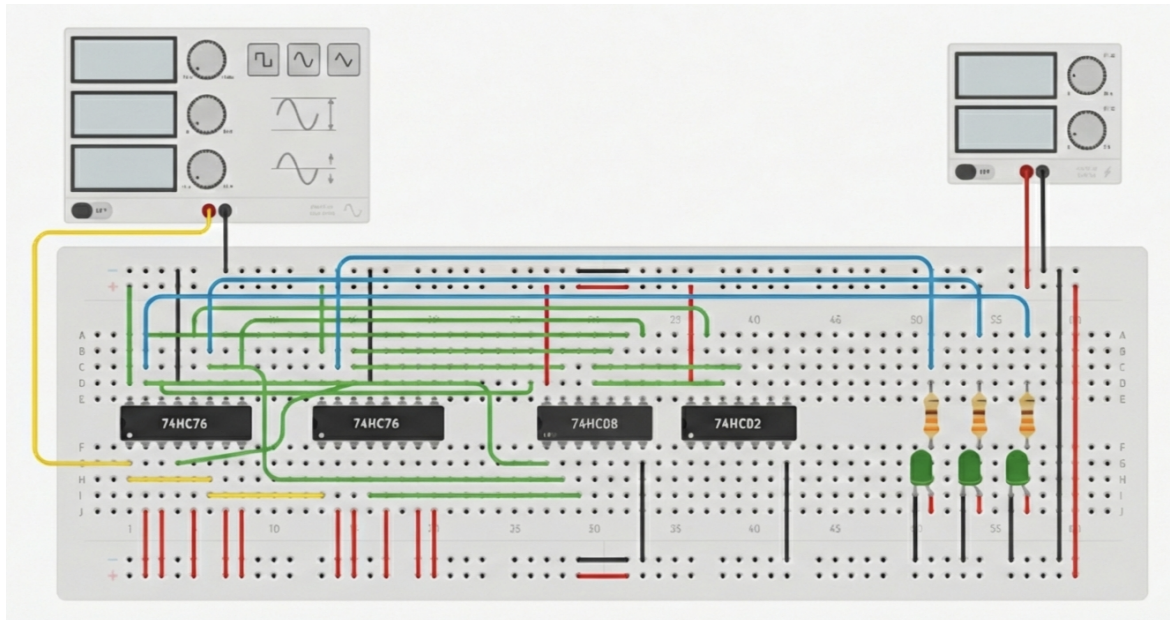
Step 5: MSB Flip-Flop (Q_2) Wiring

- $J_2 = Q_1 \cdot Q_0$:
 - Connect IC1 Pin 11 (Q_1) to AND gate IC3 Pin 2.
 - Connect IC1 Pin 15 (Q_0) to AND gate IC3 Pin 1.
 - Connect the AND output IC3 Pin 3 to IC2 Pin 4 (which is J_2).
- $K_2 = 1$: Connect IC2(Pin 16) (which is K_2) directly to +5V.

Step 6: Outputs

- Connect IC1(Pin 15) to the LSB LED (Right-most).
- Connect IC1(Pin 11) to the Middle LED.
- Connect IC2(Pin 15) to the MSB LED (Left-most).

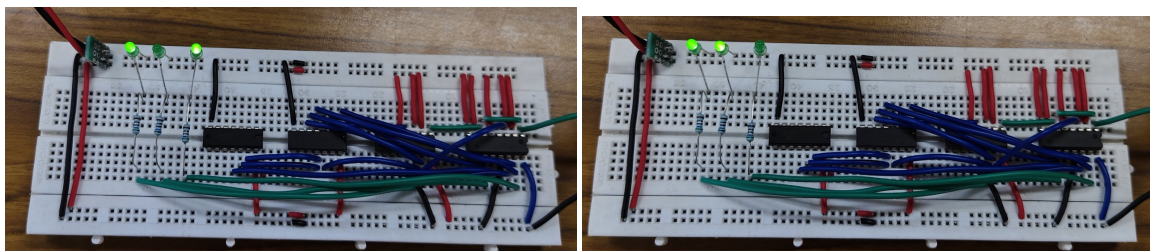
1.5 Circuit Diagram



Mod5 counter circuit

1.6 Observations

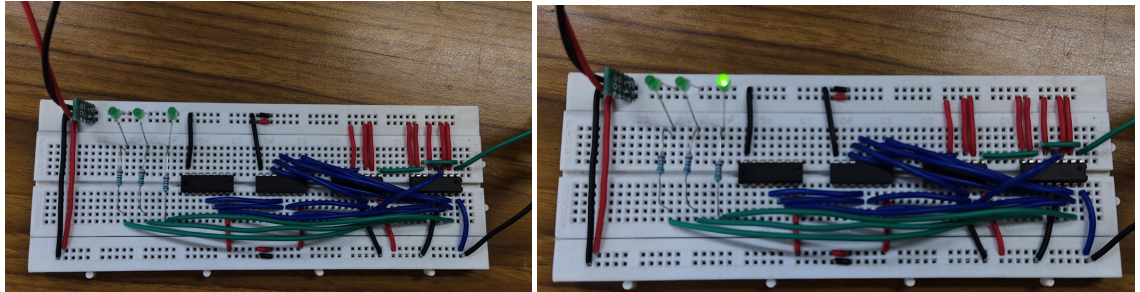
- Initially 2 to 3 times we were locked in the state 5,6 it is due to because we connected the input and output pins reversly for NOR gate



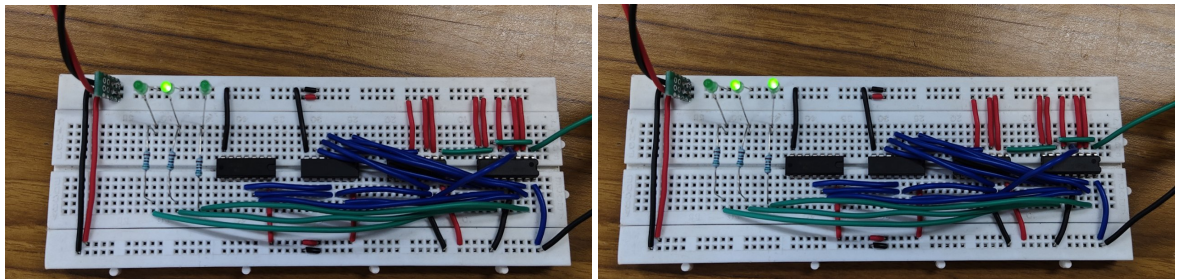
Locked between states 5 and 6

- After this we connected them properly and we got the correct output
- Clock which updates the value of the counter can be given in two ways using arduino and using AFG.
- Here we have given clock using AFG where we set the frequency of square wave to 1Hz
- When we increase the frequency of the square pulse the time taken for changing state decrease and viceversa

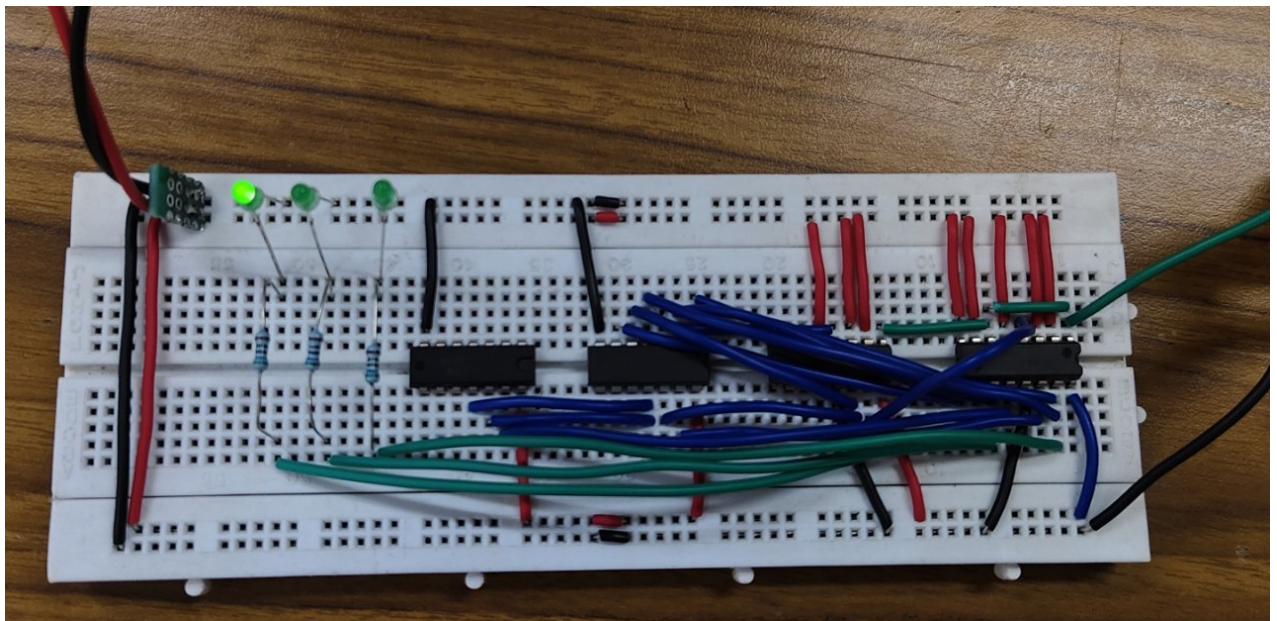
Output



Mod 5 counter with states 0 and 1



Mod 5 counter with states 2 and 3



Mod5 counter with state 4

1.7 Pin Diagram of IC's Used

